GigaWhat Energy

GigaWhat Gas
Controlled procedure

PR-GAS-031 Purging and Commissioning of Polyethylene Mains

Field	Value
Document ID	PR-GAS-031
Version	3
Effective from	2026-04-01
Review due	2029-04-01
Owner	Head of Gas Network Operations
Approved by	Gas Safety Manager
Applies to	All GigaWhat Gas sites: low pressure and medium pressure polyethylene mains

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets out how GigaWhat Gas purges new, replacement and re-laid polyethylene (PE) mains of air and commissions them with natural gas. It uses an indirect purge: air is displaced with nitrogen, then the nitrogen is displaced with natural gas, so that no flammable mixture forms inside the main.

2 Scope

This procedure applies to PE mains at low pressure (LP, up to 75 mbar) and medium pressure (MP, 75 mbar to 2 bar), such as MAIN-LP-17 (Mill Lane, 125 mm) and MAIN-MP-03 (Vale Road, 180 mm), up to 250 mm diameter and 1,000 m per purge section.

It does not cover intermediate pressure (IP) mains such as MAIN-IP-01, metallic mains, service pipes or decommissioning. A purge outside this scope needs a written purge plan approved by the Gas Safety Manager.

3 References

- PR-GAS-021 Permit-to-Work for Gas Operations
- PR-GAS-040 Working in Hazardous Areas (DSEAR Zones)
- PR-GAS-003 Response to Reported Gas Escapes
- PR-CORP-001 Reporting Incidents and Near Misses
- PR-CORP-004 Lone Working
- Gas Safety (Management) Regulations 1996 and HSE guidance L80

- Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR) and HSE guidance L138
- Pressure Systems Safety Regulations 2000 and HSE guidance L122
- HSE guidance HSG250 (permit-to-work systems) and HSG253 (safe isolation of plant and equipment)

4 Definitions

Indirect purge: A two-stage purge in which air is displaced by nitrogen and the nitrogen is then displaced by natural gas.

Vent stack: A temporary vertical metallic pipe at the far-end purge point, through which nitrogen and gas are released to atmosphere.

Gas-in-air: The concentration of natural gas in a sample, as a percentage by volume. The lower explosive limit (LEL) of natural gas is about 5% gas-in-air.

Gas-in: The moment natural gas is first admitted to the main.

Exclusion zone: The area around the vent stack that is kept clear while the vent is open, as set out in 6.2.

Purge record: Form GAS-F031, on which the purge is planned, the readings are recorded and the purge is signed off.

5 Responsibilities

Authorised Person (Gas): Approves the purge plan, issues the permit-to-work under PR-GAS-021, releases each hold point, and signs the purge record before gas-in.

Competent Person (Pressure Systems): Carries out the purge under the permit, operates the valves, takes and records all readings, and stops the purge if conditions become unsafe.

Standby person: A second trained member of the purge team (a Competent Person (Pressure Systems) or a First Call Operative) who stays at the vent stack with a dry powder extinguisher while the vent is open and keeps the exclusion zone clear.

Gas Network Controller: Is told when the purge starts, before the purge to gas and when the main is in service, and logs each event.

Gas Safety Manager: Approves any purge plan outside the scope of this procedure.

6 Procedure

6.1 Planning and authorisation

6.1.1 Prepare a purge plan on the purge record showing the main section, its length and diameter, the purge points, the vent stack location and the nitrogen volume required.

6.1.2 Confirm from the test certificate that the main has passed its strength and tightness test and that the result is recorded.

6.1.3 Calculate the nitrogen volume as at least 1.5 times the geometric volume of the main section, and arrange enough nitrogen with a pressure-regulated outlet and flow meter.

6.1.4 Obtain a permit-to-work from the Authorised Person (Gas) under PR-GAS-021 covering both the purge and the commissioning.

6.1.5 Tell the Gas Network Controller the planned start time, the main ID and the network valve that will supply gas.

6.1.6 Bump test the gas detector and the oxygen meter and confirm both are working before use.

> HOLD POINT: Do not start the nitrogen purge until the Authorised Person (Gas) has confirmed the test result, the purge plan and the permit-to-work.

6.2 Vent stack and exclusion zone

> WARNING: Vented gas can drift at low level in still or damp weather and be ignited by vehicles, people smoking or electrical equipment. A vent that is too low or an exclusion zone that is too small can lead to a flash fire.

6.2.1 Site the vent stack in open air, away from building openings, drains, overhead lines and passing traffic.

6.2.2 Fit the vent stack so that its outlet is at least 2.5 m above ground level and points vertically upwards.

6.2.3 Set out a 5 m exclusion zone around the vent stack with barriers and signs, and keep it free of ignition sources, vehicles and members of the public.

6.2.4 Earth the metallic vent stack with a driven earth spike to discharge static electricity.

6.2.5 Stop the purge and close the supply valve if any person, vehicle or ignition source enters the exclusion zone.

6.2.6 Position a standby person at the vent stack with a 9 kg dry powder extinguisher for the whole time the vent is open, during both the nitrogen purge and the purge to gas.

6.3 Nitrogen purge (air to nitrogen)

> WARNING: Nitrogen is an asphyxiant. Do not vent nitrogen into excavations, chambers or enclosed spaces, and do not lean into an excavation where nitrogen may have collected.

6.3.1 Confirm that all valves on the section are in the positions shown on the purge plan and that the vent valve is closed.

6.3.2 Connect the nitrogen supply to the inlet purge point through the regulator and flow meter, and set the flow rate shown on the purge plan.

6.3.3 Open the vent valve fully, then open the nitrogen supply valve slowly.

6.3.4 Start taking oxygen readings at the vent sample point when 75% of the calculated nitrogen volume has been admitted, and take a reading every minute.

6.3.5 Continue the nitrogen purge until the oxygen reading at the vent is 3% or less.

6.3.6 Close the vent valve, then close the nitrogen supply valve, leaving the main holding nitrogen at between 20 and 50 mbar.

6.3.7 Record the nitrogen volume used, the final oxygen reading and the time on the purge record.

6.3.8 Give the completed nitrogen section of the purge record to the Authorised Person (Gas) for review.

> HOLD POINT: Do not admit gas to the main until the Authorised Person (Gas) has reviewed the nitrogen purge results and signed the purge record to authorise gas-in.

6.4 Purge to gas (nitrogen to natural gas)

> WARNING: Once gas reaches the vent, the mixture leaving the vent passes through the flammable range. Keep all ignition sources out of the exclusion zone and never leave an open vent unattended.

6.4.1 Tell the Gas Network Controller that the purge to gas is about to start.

6.4.2 Open the vent valve fully.

6.4.3 Open the gas supply valve slowly, a quarter turn at a time, and control the flow at the rate shown on the purge plan.

6.4.4 Start taking gas-in-air readings at the vent sample point as soon as gas is first detected, and take a reading every minute.

6.4.5 Treat the purge as complete only when three consecutive readings of at least 90% gas-in-air, taken one minute apart, have been obtained with a gas detector calibrated within the last 6 months.

6.4.6 Close the vent valve and record the readings and times on the purge record.

6.4.7 Leave the vent stack in place until commissioning is complete.

6.5 Commissioning and return to service

6.5.1 Raise the main slowly to its operating pressure by opening the gas supply valve fully.

6.5.2 Test all purge point fittings, and any joints made during the purge, with a gas detector and leak detection fluid.

6.5.3 Remove the vent stack, cap the purge points and check them gas-tight.

6.5.4 Hand the completed purge record to the Authorised Person (Gas).

> HOLD POINT: Do not put the main into service or connect any customer until the Authorised Person (Gas) has confirmed that the main is at operating pressure and gas-tight, and has cancelled the permit-to-work.

6.5.5 Tell the Gas Network Controller that the main is in service and give the time.

6.5.6 Commission service connections only after the main has been confirmed live and gas-tight.

6.6 Emergency

> WARNING: Do not try to put out a gas flame at the vent while gas is still flowing. Unburnt gas can collect and re-ignite with greater force.

6.6.1 If gas ignites at the vent, close the gas supply valve and let the flame burn out at the vent.

6.6.2 If anyone is injured, or there is fire beyond the vent, call 999.

6.6.3 If gas escapes uncontrollably, close the supply valve, clear people from the area and report the escape to the Gas Network Controller for handling under PR-GAS-003, or call the National Gas Emergency Service on 0800 111 999 if the Controller cannot be reached.

6.6.4 Report any ignition, uncontrolled release or near miss under PR-CORP-001.

7 Records

The completed purge record (form GAS-F031), with readings, times and signatures, is kept with the permit-to-work in the asset file for the life of the main.

The Gas Network Controller's log records the start of the purge, the purge to gas and the time the main went into service. The asset register is updated with the commissioning date.

8 Revision history

Version	Date	Change
1	2018-03-05	First issue.
2	2022-07-01	Indirect (nitrogen) purge made mandatory for all PE mains in scope. Added commissioning hold point and emergency section.
3	2026-04-01	Revised after incident INC-2025-031 (vented gas drifted towards a parked vehicle during a purge of MAIN-MP-03). Purge to gas complete only after three consecutive readings of at least 90% gas-in-air, one minute apart, with a detector calibrated within the last 6 months (was a single reading of at least 90% gas). Vent stack raised from 2 m to 2.5 m; exclusion zone increased from 3 m to 5 m and now kept free of vehicles. Added a standby person with a dry powder extinguisher at the vent, and a hold point for the Authorised Person (Gas) to sign the purge record before gas-in.